Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	i		Some water used by {photosynthesis/metabolic reactions}/ water produced in respiration/ measures the rate of absorption not the rate of transpiration		1		1		
		ii		**Any 3 (x1) from:** humidity/ or description (1) wind/ air currents (1) surface area of leaves (1) age of leaves (1) Accept air pressure NOT same number/ mass of leaves/ length of stem/ plant	2	1		3		3
	(b)			Lower surface of oak leaf shaded/ or description of/ ORA (1) so higher density of stomata to reduce water loss (1) OR neither surface of wheat shaded/ or description of (1) equal distribution of stomata water loss equal both sides (1)		2		2		

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
	(c)			1. (Potassium ions/ malate) reduce water potential in (guard) cell (1) Accept osmotic pressure increases/ osmotic potential decreases/ solute potential decreases/ hypertonic to outside 2. water moves in by osmosis (down water potential gradient) (1) 3. (Turgor) pressure inside (guard) cell increases/ cells become turgid (1) NOT cells expand 4. ends of guard cell have a thinner wall than centre/ ORA (1) 5. ends of guard cell expand and stomata opens (1)	3		2	5		
	(d)	i		2 x π x 2 x 423 = 5312.9 (1) = 5310 (to 3 sig figs) (1) Allow 5320 if they use value of π from calculator. 5310/ 5320 = 2 marks 5312.9/ 5313/ 5315.6/ 5316 = 1 mark Evidence of 2 πr x 423 = 1 mark		2		2	2	
		ii		(water molecules) escape more readily from) species B because it has larger (total) circumference. Ecf if calculation incorrect in (i)		1		1		
				Question 4 total	**5**	**7**	**2**	**14**	**2**	**3**